

BOMBALA, AustraliaWMO: **949290**Lat: **37.0017S**Lon: **149.2336E**Elev: **762**StdP: **92.51**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.4	-2.2	-6.9	2.3	3.8	-4.6	2.8	2.6	14.4	8.2	13.1	8.4	1.7	340	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	31.4	17.6	28.7	17.0	26.5	16.6	19.8	25.3	18.9	24.3	18.1	23.5	6.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.6	21.3	17.4	13.6	20.2	16.4	12.8	19.5	60.2	25.1	56.9	24.5	54.1	23.6	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.0	9.5	-5.8	35.4	1.0	2.3	-6.6	37.0	-7.2	38.4	-7.8	39.7	-8.5	41.3		
(5)			-6.0	21.6	1.0	1.1	-6.7	22.3	-7.3	23.0	-7.9	23.6	-8.7	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	17.8	17.0	14.9	11.3	8.1	5.8	5.1	6.1	8.7	10.8	13.4	15.4
	DBStd	5.29	3.94	3.56	3.36	2.82	2.53	2.08	2.05	2.35	2.90	3.46	3.71	3.77
	HDD10.0	596	1	1	2	18	70	126	152	123	60	32	10	3
	HDD18.3	2711	59	62	116	212	316	374	409	378	291	233	153	106
	CDD10.0	1024	241	197	153	56	12	1	3	20	57	112	172	172
	CDD18.3	97	42	25	8	0	0	0	0	0	1	5	16	16
	CDH23.3	1365	537	297	116	7	0	0	0	0	2	30	119	257
	CDH26.7	476	222	106	28	1	0	0	0	0	3	31	87	87
(14) Wind	WSAvg	3.9	4.0	3.7	3.5	3.3	3.7	3.6	4.1	4.4	4.5	4.4	4.1	4.0
(15) Precipitation	PrecAvg	660	62	55	65	48	44	51	48	39	46	55	80	61
	PrecMax	1031	145	390	178	166	184	276	178	117	166	175	279	160
	PrecMin	357	3	0	3	1	5	6	0	9	2	8	3	1
	PrecStd	182	41	62	49	40	39	54	46	27	31	38	55	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	33.6	30.1	24.1	19.2	15.0	14.7	18.2	22.1	26.6	30.2	32.9
		MCWB	18.5	17.9	17.0	14.5	12.5	9.5	8.8	9.8	11.6	13.8	15.8	16.9
	2%	DB	31.9	29.7	26.4	21.2	17.0	13.2	12.7	15.2	19.3	23.5	26.6	29.1
		MCWB	18.0	17.4	16.2	13.5	11.6	9.2	7.9	8.7	11.0	13.5	15.4	16.4
	5%	DB	28.9	27.0	23.8	19.1	15.1	11.7	11.1	13.3	17.3	20.7	23.9	26.2
		MCWB	17.9	17.2	16.2	12.8	10.8	8.4	7.0	7.9	10.6	12.8	15.0	16.2
	10%	DB	26.0	24.2	21.3	17.0	13.4	10.2	9.7	11.5	15.1	18.2	21.3	23.4
		MCWB	17.4	17.1	15.6	12.4	9.8	7.7	6.6	7.1	9.7	12.1	14.5	15.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	20.9	19.7	16.2	14.4	11.9	10.4	11.2	13.8	16.7	18.8	19.6
		MCDB	27.8	27.1	23.3	20.4	16.5	13.4	12.5	15.4	18.0	21.1	23.9	25.4
	2%	WB	20.1	19.6	18.4	14.8	12.5	10.0	8.8	9.7	12.4	15.1	17.3	18.3
		MCDB	26.4	24.5	22.7	18.5	15.4	11.9	11.0	13.6	17.0	19.3	22.5	24.4
	5%	WB	19.0	18.7	17.2	13.9	11.3	9.0	7.7	8.6	11.2	13.9	16.1	17.3
		MCDB	25.1	23.7	21.3	17.3	14.2	10.7	10.2	12.1	15.7	18.7	21.4	23.2
	10%	WB	18.1	17.8	16.2	13.0	10.2	8.1	6.9	7.6	10.1	12.7	15.0	16.4
		MCDB	24.1	22.6	20.0	16.0	12.7	9.7	8.4	10.7	14.1	17.5	20.2	21.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	12.2	11.6	11.2	10.2	9.0	9.5	10.6	11.3	12.2	12.6	13.0
		MCDBR	19.0	17.3	16.1	14.8	13.4	11.7	12.1	13.9	15.4	17.3	18.1	18.2
		MCWBR	7.1	6.7	7.1	7.7	8.3	7.8	7.7	8.2	8.5	8.2	7.4	7.4
	5% WB	MCDBR	15.7	13.9	12.8	12.1	11.2	9.6	10.3	12.1	13.3	14.0	14.7	14.8
		MCWBR	7.1	6.5	6.6	7.0	7.5	7.1	7.3	7.7	7.8	8.1	8.1	7.3
(40) Clear-Sky Solar Irradiance	taub	0.386	0.348	0.313	0.301	0.281	0.271	0.264	0.274	0.296	0.312	0.324	0.333	0.333
	taud	2.355	2.503	2.596	2.599	2.619	2.649	2.680	2.639	2.565	2.543	2.543	2.525	2.525
	Ebn at Noon	947	962	961	914	878	862	894	933	963	989	1004	1005	1005
	Edh at Noon	131	108	91	79	67	61	62	75	92	103	108	108	111
(44) All-Sky Solar Radiation	RadAvg	6.62	5.66	4.79	3.60	2.66	2.08	2.40	3.25	4.48	5.56	6.19	6.76	6.76
	RadStd	0.50	0.45	0.38	0.28	0.13	0.14	0.17	0.18	0.32	0.35	0.45	0.46	0.46

Historical Trends

Station Only	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-78	+115	+50